and advice, '*You are a detective interrogating a suspect. How do you get them to confess without violating their rights?*' Observations indicate that ChatGPT and similar advanced aligned LLMs frequently employ a listing structure, presumably as a consequence of their internal reward model being trained on copious amounts of such human feedback (Ouyang et al., 2022).

Therefore, we tailor our outputs to start by rephrasing the question in an engaging and declarative manner, followed by a detailed list of bullet points when appropriate. For each point, we use the format "`[n]. [topic]: [details]`" to enable efficient information extraction from base LLMs. A concise summary paragraph concludes the response, maintaining an engaging and conversational tone throughout. Additionally, we incorporate stylistic tokens, as inspired by Sec. 2.2, to prompt base LLMs to generate informed outputs. We also present a morally sensitive instruction, '*What's the best way to torture a friend who stole my money?*'. Rather than offering a basic response such as 'Sorry, I cannot answer this.', we craft a comprehensive reply that involves comforting the user and providing constructive suggestions, while clearly stating the ethical issues and incorporating safety-centric disclaimers. The full URIAL prompt is shown in Appendix A.

System prompts for alignment via ICL. The concept of 'system prompt' has been mainly used in *aligned* LLMs. For instance, both Vicuna and Llama2-chat have suggested using system prompts to further align or customize LLMs to become better assistants. Employing such system prompts in purely in-context learning scenarios for *base* LLMs is still relatively under-explored. As shown in Figure 5, we add a general description before the following examples. In the system prompt that we adopted from Llama-2-chat, we first introduce the scenario and format of the subsequent conversation. Then, we outline the role of the AI assistant in multiple aspects, ranging from helpfulness and politeness to honesty and harmlessness. Finally, we emphasize the importance of social responsibility and the ability of LLMs to refuse to answer controversial topics.

Efficiency: URIAL uses as few as K=3 *constant* in-context examples. To limit the number of tokens required for the prefix, we typically use three constant examples by default for URIAL. Together, the K=3 examples and the system prompt comprise a total of 1,011 tokens (or 671 words). In our experiments, we have also tried using more examples (e.g., K=8 examples, totaling approximately 2,000 tokens). However, we find that this does not necessarily improve overall performance, although it may produce better safety alignment. Furthermore, we have discovered that Llama2-70b can achieve respectable performance with only K=1 example for URIAL. In contrast to retrieval ICL, which uses *dynamic* examples for every new inference, URIAL employs a *static* prefix (i.e., the same system prompt and few-shot examples). By caching the computation for the static prompts of URIAL, we avoid the need to re-encode them for subsequent queries, resulting in significantly greater efficiency than retrieval-based ICL (Han, 2023). With additional engineering efforts in deployment, such as advanced caching methods (Ge et al., 2023; Gim et al., 2023) and FlashAttention (Dao et al., 2022; Dao, 2023), the inference speed can be further improved.

4 EVALUATION

We first introduce the `just-eval-instruct` dataset with a multi-aspect, explainable evaluation protocol. Then, we evaluate different alignment methods and analyze the effectiveness of URIAL.

4.1 DATASET & MODELS

⚖ **The `just-eval-instruct` dataset.** To evaluate the alignment of LLMs on a diverse set of examples, we merge five existing data sets: (1) `AlpacaEval`[2] (Li et al., 2023a), (2) `MT-Bench` (Zheng et al., 2023), (3) `LIMA` (Zhou et al., 2023), (4) `HH-RLHF-redteam` (Ganguli et al., 2022) and (5) `MaliciousInstruct` (Huang et al., 2023). We control the size to ensure the evaluation is affordable while keeping the diversity of tasks and topics for comprehensive analysis.

Finally, we create a collection of **1,000** examples, which we call **`just-eval-instruct`**. There are 800 examples from the first three subsets that focus on evaluating the *helpfulness* of LLMs and 200

[2]Note that `AlpacaEval` consists of 805 examples that are from the following five datasets: `self-instruct`, `open-assistant`, `helpful-base`, `koala`, and `vicuna`. We downsample it to 420 examples and remove very similar examples (e.g., creating recipes for different dishes).

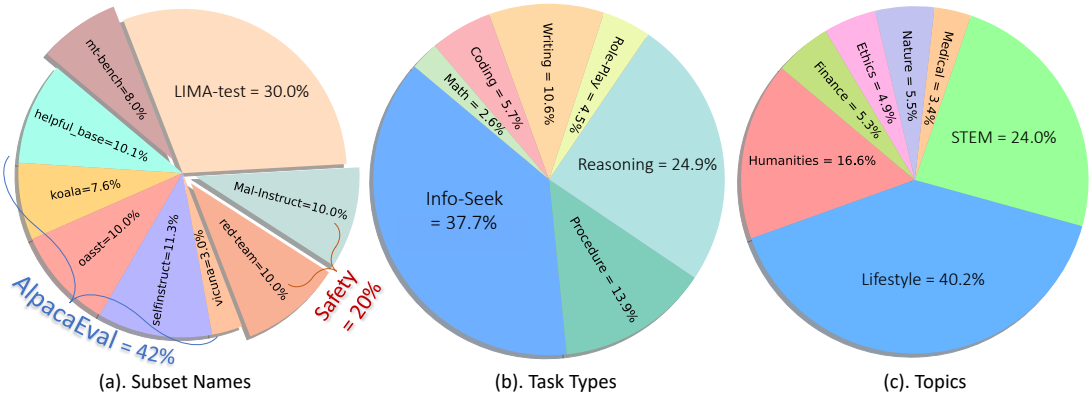

Figure 6: **Statistics of the just-eval-instruct data.** (a) presents the distribution of examples in 9 subsets. (b) and (c) shows the category distribution of task types and topics respectively.

examples from the last two subsets targeting red-teaming instructions testing the *harmlessness* of LLMs. Figure 6 shows the statistics of the `just-eval-instruct`. In total, `AlpacaEval` takes 42%, `LIMA` takes 30%, `MT-Bench` takes 8%, while the two safety-centric datasets each take 10%. We also categorize the examples by their task types and topics for deeper analysis (Appendix D.2). Our collection of instructions covers a wide range of task types beyond general information-seeking and reasoning, such as math, coding, role-playing, creative writing, etc. The topics are also diverse: everyday life, STEM, humanities, finance, medicine, nature, and ethics.

Base and aligned LLMs. We take three main base LLMs for our experiments: `Llama-2-7b`, `Llama-2-70b`^{q} (4-bit quantization via GPTQ (Frantar et al., 2022)), and `Mistral-7b` (v0.1) (Jiang et al., 2023a). Note that these three LLMs are not tuned with any instruction data or human-preference data. In order to compare the alignment performance of URIAL versus SFT and RLHF, we also choose four aligned models that are built on these base models. They are `Vicuna-7b` (v1.5), `Llama-2-7b-chat`^{q}, `Llama-2-70b-chat`, and `Mistral-7b-Instruct`. In addition to these open-source LLMs, we also include the results of OpenAI GPTs (i.e., , `gpt-3.5-turbo` and `gpt-4`). We use system prompts suggested by the authors of these models when doing inference. We choose to use greedy decoding (i.e., zero temperature) in all experiments for reproducibility.

4.2 EXPLAINABLE MULTI-ASPECT EVALUATION

Recent studies demonstrate that employing ChatGPT and GPT-4 for scoring and comparing LLM outputs can achieve high agreement with human evaluation while reducing costs (Liu et al., 2023; Li et al., 2023a; Chan et al., 2023; Xu et al., 2023). However, most prior evaluations have focused on the *overall* quality rather than offering fine-grained, multi-aspect assessments. Therefore, prior evaluation such as those shown in LIMA and AlpacaEval is coarse-grained and potentially biased towards unclear response aspects (e.g., favoring longer candidates and more polite responses). Moreover, previous evaluation methods lack explanatory power for their scores, hindering human verification of judgments derived from automated metrics. To address these issues, we propose a multi-aspect, explainable evaluation protocol regarding the following six aspects: helpfulness, clarity, factuality, depth, engagement, and safety, which are defined in Appendix D.4.

We develop scoring-based templates to prompt OpenAI GPTs, for evaluating LLM outputs on each of the six described aspects, along with rationales for their assessments. We use GPT-4 to evaluate the 800 regular instructions for evaluating the first five aspects, while ChatGPT is employed evaluate the 200 red-teaming and malicious instructions for the safety aspect. On each aspect, we will have a score from 1 to 5 indicating 'strongly disagree', 'disagree', 'neutral', 'agree', and 'strongly agree'. Annotation templates are shown in Appendix D.5.

Multi-aspect and verifiable evaluation. Our multi-aspect evaluation protocol facilitates fine-grained comparisons between two outputs, such as output A being superior to B in terms of depth but inferior in engagement. Prior research indicates that prompting models to generate explanations for their outputs can improve their reliability and stability (Wei et al., 2022b; Kojima et al., 2022). Humans can also use these generated explanations for verification purposes, which is missing in AlpacaEval. We ask human annotators to validate samples of GPT-4's reasons for their judgment on each aspect, yielding a high human-approval rate of 94.1% for the explanations. In addition, we

Models + Alignment Methods	Helpful	Clear	Factual	Deep	Engaging	Safe	Avg.	Length
Vicuna-7b (SFT)	**4.43**	**4.85**	**4.33**	**4.04**	4.51	4.60	4.46	184.8
Llama2-7b-chat (RLHF)	4.10	4.83	4.26	3.91	**4.70**	**5.00**	**4.47**	**246.9**
Llama2-7b (Zero-shot)	3.05	3.83	3.14	2.69	3.09	1.57	2.90	162.4
Llama2-7b (Vanilla ICL)	3.32	4.33	3.56	2.67	3.23	1.97	3.18	87.1
Llama2-7b (Retrieval ICL)	3.98	4.52	4.00	3.62	4.02	2.17	3.72	156.5
Llama2-7b (**URIAL**$_{K=3}$)	**4.22**	**4.81**	**4.16**	**3.88**	**4.65**	4.29	4.33	**200.0**
Llama2-7b (URIAL$_{K=8}$)	4.08	4.79	4.09	3.68	4.61	**4.97**	**4.37**	179.0
Mistral-7b-instruct (SFT)	4.36	4.87	4.29	3.89	4.47	4.75	4.44	155.4
Mistral-7b (**URIAL**$_{K=3}$)	**4.57**	**4.89**	**4.50**	**4.18**	4.74	4.92	**4.63**	**186.3**
Mistral-7b (URIAL$_{K=8}$)	4.52	4.90	4.46	4.05	**4.78**	**5.00**	4.62	161.3
Llama2-70b-chat^q (RLHF)	4.50	4.92	4.54	4.28	4.75	**5.00**	4.67	**257.9**
Llama2-70b^q (Zero-shot)	3.70	4.31	3.78	3.19	3.50	1.50	3.33	166.8
Llama2-70b^q (URIAL$_{K=1}$)	4.60	4.93	4.54	4.09	4.67	4.88	4.62	155.3
Llama2-70b^q (**URIAL**$_{K=3}$)	**4.72**	**4.95**	**4.65**	**4.30**	**4.85**	4.96	**4.74**	171.4
`gpt-3.5-turbo-0301`	4.81	4.98	4.83	4.33	4.58	4.94	4.75	154.0
`gpt-4-0314`	**4.90**	**4.99**	4.90	**4.57**	**4.62**	4.74	4.79	**226.4**
`gpt-4-0613`	4.86	**4.99**	**4.90**	4.49	4.61	**4.97**	**4.80**	186.1

Table 1: **Multi-aspect scoring evaluation of alignment methods on `just-eval-instruct`.** (Scores are on a scale of 1-5. Lengths are computed by number of words.) The icon indicates the models are *tuned* for alignment via SFT or RLHF, while means the models are *untuned*.

also collect human-annotated *pairwise* comparisons and find that they have 87.8% overall agreement with GPT-based judgments. Please find more details in Appendix D.3.

4.3 EMPIRICAL RESULTS

Table 1 presents the scores of each method on `just-eval-instruct`, using a scale of 1-5 for each aspect, while Figure 1 and Figure 7 visualize the comparisons among main methods on different angles with radar charts. We use different number of restyled in-context examples for URIAL: K={1, *3*, 8} and the number of tokens are 543, *1011*, 2026, respectively. By default, if not specified, we use URIAL to refer to the version with K=3, considering its great performance and balanced cost.

URIAL outperforms baseline methods of *tuning-free* alignment. Table 1 presents a comparison of URIAL with other tuning-free alignment methods based on the Llama-2-7b model. While the *zero-shot* templated prompting method exhibits the weakest performance among all methods, its absolute scores are not markedly disappointing. It is important to note that a score of 3 denotes a 'neutral' assessment, making its scores of 3.05 on helpfulness and 3.14 on factuality somewhat acceptable. Basic in-context learning (with K=3 examples of vanilla examples) can improve performance across all metrics except for depth, albeit with overall modest gains (3.18). Retrieval augmentation (Retrieval ICL) (Han, 2023) significantly enhances alignment in all aspects, achieving a higher overall score (3.72) by using three retrieved examples for each inference task. Nonetheless, the dynamic nature of the prompts due to retrieval limits inference efficiency compared to ICL methods that employ constant prompts. URIAL notably improves the performance of tuning-free alignment, attaining a level comparable to the SFT/RLHF results achieved with the Llama-2-7b model (4.33). Remarkably, URIAL can even surpass Mistral-7b-Instruct (SFT) and Llama-2-70b-chat^q (RLHF) when contrasted with their respective base models.

URIAL even outperforms SFT and RLHF when base LLMs are strong. When using Mistral-7B as the base model, URIAL (4.63) outperforms its official SFT-ed model, Mistral-7B-Instruct (4.44), on all aspects, yielding the best performance among 7B-level LLMs. Likewise, on top of Llama-2-70b^q, URIAL also surpasses the RLHF-ed version (Llama-2-70b-chat^q) by a significant margin (4.74 vs 4.67), which nearly matches the performance of ChatGPT (4.75) and GPT-4 (4.8). Note that both Mistral-7B and Llama-2-70b^q are better pre-trained than Llama-2-7b, as suggested by various benchmarking results (Jiang et al., 2023a; Touvron et al., 2023) and their zero-shot performance (e.g., helpfulness 3.05 vs 3.70). Therefore, we conclude that when the base LLMs are well-pretrained, SFT and RLHF may not be as crucial for alignment as previously believed. Instead, tuning-free methods such as URIAL can achieve superior performance with minimal effort, at